

DISTRIBUTED SYSTEMS

Essay Type Questions (ETQ)

30 Comprehensive Scenario-Based Questions

Each Question: 10 Marks (5 sub-questions × 2 marks)

This document contains 30 theory-type essay questions designed for examining understanding of distributed systems concepts. Each question follows a consistent structure: a realistic scenario (often set in a university or institutional context) followed by five analytical sub-questions (a–e). The questions cover key topics including hybrid architectures, decoupling, event-driven systems, commit protocols, Byzantine fault tolerance, RPC, middleware, naming, P2P systems, scalability, coordination services, logical time, and resilience patterns.

Instructions for Students: Use the provided source context and theoretical knowledge to develop comprehensive, well-structured responses. Each sub-question carries equal weight (2 marks).

Topics Covered: Hybrid architectures • Space/Time/Sync decoupling • Event Notification vs Carried State Transfer • Message brokers & Pub-Sub • Commit protocols (2PC/3PC) • Byzantine faults & Rule of Tolerance • Two Generals problem • Resilience patterns (timeouts, retries, circuit breakers) • Saga pattern • Monolith to microservices • RPC anatomy & transparency • Communication paradigms • Cascading faults • Apache Kafka • RMI vs RPC • Persistent vs transient messaging • IBM MQ • Integration buses • Failure taxonomies • Impossibility results • Failure masking • Design for Recovery • Middleware pipelines • Layered consistency • DNS, LDAP & Hilbert curves • P2P structured/unstructured • Scalability triad • Mutual exclusion evolution • Snapshots (Chandy-Lamport) • Leader election (Raft/ZooKeeper) • Logical vs physical time • Vector clocks • Spanner TrueTime • Lamport causality.

ETQ 1 (10 Marks): Hybrid Network Architecture and Resource Management

Scenario:

A private university in Benin City keeps student records, staff email and backups on central servers. However, authorised research teams exchange large datasets directly between their workstations. During a server failure, the research teams can still share files, but the registry and email services become unavailable.

- a. (2 marks)** Analyse the network architecture used by the university and justify your conclusion from the way resources are provided.
- b. (2 marks)** Differentiate the roles of the centralised and peer-based parts of the network, and explain the weakness revealed by the outage.
- c. (2 marks)** Evaluate how retaining the hybrid architecture can support future growth, while also creating additional management responsibilities for the ICT unit.
- d. (2 marks)** Discuss the security and data consistency risks that arise when research teams share large datasets outside central governance.
- e. (2 marks)** Propose strategies the ICT unit could adopt to improve resilience of central services while preserving the benefits of peer-to-peer research collaboration.

ETQ 2 (10 Marks): Dimensions of Decoupling in Event-Driven Campus Notification Systems

Scenario:

The university administration wants to notify students and staff about lecture cancellations, venue changes and emergency closures. The current system uses direct synchronous API calls from the central timetable database to email, SMS and mobile push services. This causes timeouts and lost notifications during peak periods such as exam registration. The IT team is evaluating a publish-subscribe model with a message broker.

- a. (2 marks)** Analyse how Space, Time and Sync decoupling collectively transform interaction between the timetable producer and notification consumers.
- b. (2 marks)** Differentiate Event Notification versus Carried State Transfer in this notification system, focusing on bandwidth, autonomy and complexity.
- c. (2 marks)** Detail the operational workflow of the message broker from publication of a cancellation event to final delivery to subscribers.
- d. (2 marks)** Evaluate why debugging, ordering and duplicate messages are non-trivial challenges in this Pub-Sub implementation at scale.
- e. (2 marks)** Justify why Pub-Sub is better suited for university alert systems than traditional synchronous request/response, referencing examples such as weather updates or live chat.

ETQ 3 (10 Marks): Architectural Trade-offs in Event Delivery for Student Fee and Result Notifications

Scenario:

A university's finance and examinations offices need to notify thousands of students about fee payments due and result releases. Direct synchronous calls from the central database frequently overload the email and SMS gateways. The team is considering an event-driven approach where the finance/exam systems publish events and independent services handle delivery.

- a. (2 marks)** Compare the bandwidth and autonomy implications of Event Notification versus Carried State Transfer for high-volume student notifications.
- b. (2 marks)** Analyse how the message broker reduces coupling while introducing new failure modes.
- c. (2 marks)** Discuss the advantages and disadvantages of each event delivery style for a system that must guarantee delivery of fee and result information.
- d. (2 marks)** Evaluate the impact of each style on overall system complexity and operational monitoring.
- e. (2 marks)** Recommend the most suitable event delivery approach for this university scenario and justify your choice.

ETQ 4 (10 Marks): The Role of the Broker in a University Lost-and-Found and Event Publication System

Scenario:

The university operates a central 'lost and found' service and an events calendar. Departments currently call the central service directly to publish or query items. The IT team plans to introduce a message broker so that any department can publish 'item found' or 'event scheduled' messages without knowing which other services (student portal, security dashboard, email) will consume them.

- a. (2 marks) Detail the step-by-step operational workflow of the message broker from publication to delivery.
- b. (2 marks) Explain how the broker enables loose coupling between publishing departments and consuming services.
- c. (2 marks) Analyse how the broker handles message persistence, routing and delivery guarantees.
- d. (2 marks) Discuss the monitoring and debugging challenges introduced by asynchronous broker-mediated communication.
- e. (2 marks) Evaluate the benefits of the broker model over direct synchronous calls for this multi-department scenario.

ETQ 5 (10 Marks): Challenges of Asynchrony at Scale in a Distributed Student Discussion and Alert Platform

Scenario:

A popular university-wide discussion forum and alert platform uses a Pub-Sub backbone. During semester registration and exam result release periods the system experiences high message volume, occasional duplicate alerts and out-of-order updates visible to students.

- a. (2 marks) Identify the specific Pub-Sub challenges of debugging, ordering and duplicate messages in this high-scale student platform.
- b. (2 marks) Explain why these challenges are considered 'non-trivial' to solve in production.
- c. (2 marks) Analyse the trade-offs between stronger ordering guarantees and system throughput/latency.
- d. (2 marks) Discuss techniques (idempotency, deduplication, sequencing) that could mitigate the observed problems.
- e. (2 marks) Evaluate the overall suitability of Pub-Sub for a student-facing discussion and alert system given these challenges.

ETQ 6 (10 Marks): Trade-offs of Commit Protocols for Distributed Student Record Updates

Scenario:

A multi-campus university maintains student academic records across several departmental databases. When a student changes programme or a grade is updated, the change must be applied consistently across all relevant databases. The current two-phase commit implementation sometimes blocks for long periods during network issues.

- a. (2 marks) Compare and contrast Two-Phase Commit (2PC) and Three-Phase Commit (3PC) in the context of student record updates.
- b. (2 marks) Analyse their differences regarding network overhead and failure risk.
- c. (2 marks) Discuss the suitability of each protocol for internal centralised workflows versus truly distributed multi-campus workflows.
- d. (2 marks) Evaluate why 2PC can become a bottleneck in a university environment with variable network quality.
- e. (2 marks) Recommend whether the university should migrate to 3PC or adopt an alternative (e.g., Saga) and justify your answer.

ETQ 7 (10 Marks): Navigating Byzantine Behaviour in a Multi-Department Research Data Collaboration Platform

Scenario:

Several university research groups and external partners share experimental datasets and vote on research conclusions through a distributed platform. Some participant nodes occasionally send conflicting or fabricated data due to software bugs or compromise.

- a. (2 marks)** Analyse the challenges posed by Byzantine node behaviour where actors may actively lie or collude.
- b. (2 marks)** Explain how the 'Rule of Tolerance' provides a mathematical framework for building trust in this environment.
- c. (2 marks)** Differentiate benign (clean/omission) failures from arbitrary (malicious/corrupt) failures and the defensive architectures each requires.
- d. (2 marks)** Discuss why perfect agreement remains impossible and how 'consistency shields' can still provide practical safety.
- e. (2 marks)** Recommend a combination of timeouts, cryptographic techniques and replication strategies suitable for this research collaboration platform.

ETQ 8 (10 Marks): The Illusion of Certainty – Applying the Two Generals Problem to Distributed Exam Result Publication

Scenario:

The examinations office publishes final results to student portals, email and SMS. Network partitions between the central results database and the notification services have occasionally caused some students to see results while others do not, leading to confusion and complaints.

- a. (2 marks)** Using the Two Generals problem as baseline, analyse why perfect certainty of result delivery is impossible.
- b. (2 marks)** Explain how this reality forces engineers to design for 'permanent doubt'.
- c. (2 marks)** Discuss the role of consistency shields and acknowledgements in managing uncertainty.
- d. (2 marks)** Evaluate the impact of these impossibility results on service-level agreements for exam result publication.
- e. (2 marks)** Propose practical design patterns the university can adopt to minimise student confusion during partial failures.

ETQ 9 (10 Marks): Resilience through Strategy for Achieving High Availability in the University Portal

Scenario:

The main student and staff portal must remain available even when individual backend services fail. The current system experiences cascading outages when one microservice slows down. The ICT team is implementing timeouts, retries with exponential backoff, circuit breakers and hard limits.

- a. (2 marks)** Evaluate the roles of timeouts, retries and hard limits in managing network uncertainty.
- b. (2 marks)** Explain why engineered fallbacks are essential for achieving 'Five Nines' availability on unreliable hardware and networks.
- c. (2 marks)** Analyse how circuit breakers prevent cascading faults in the portal architecture.
- d. (2 marks)** Discuss the trade-off between aggressive retry policies and system stability.
- e. (2 marks)** Recommend a layered resilience strategy for the university portal that balances user experience and resource protection.

ETQ 10 (10 Marks): Evolution of State Management – From ACID to Saga in Microservices Course Registration

Scenario:

A university is migrating its monolithic course registration system to microservices. The new system must handle course capacity checks, fee payment, timetable updates and notification across independent services. Strict ACID transactions are causing long locks and poor scalability.

- a. (2 marks)** Explore the transition from strict ACID principles in centralised systems to the Saga pattern in modern microservices.
- b. (2 marks)** Analyse the compensating transaction mechanism in a course registration Saga.
- c. (2 marks)** Discuss the consistency guarantees provided by Saga versus traditional ACID.
- d. (2 marks)** Evaluate the operational complexity introduced by Saga choreography or orchestration.
- e. (2 marks)** Recommend whether the university should adopt Saga for registration and justify the choice.

ETQ 11 (10 Marks): The Evolution of Enterprise Software: Monolith to Microservices in University ERP

Scenario:

The university's central ERP system (student records, finance, HR, timetabling) was originally a large monolithic application. Frequent updates caused downtime for the entire institution. The ICT team is considering breaking it into independent microservices aligned with business capabilities.

- a. (2 marks)** Discuss the transition from monolithic architectures to specialized, independent services and why the 'Monolith' is described as dead in this context.
- b. (2 marks)** Analyse the new responsibilities this transition places on the university architect regarding service boundaries and data ownership.
- c. (2 marks)** Evaluate the impact on deployment frequency, team autonomy and technology diversity.
- d. (2 marks)** Discuss the challenges of distributed transactions and data consistency across the new microservices.
- e. (2 marks)** Recommend a migration strategy (strangler fig pattern, etc.) suitable for a live university ERP system.

ETQ 12 (10 Marks): Navigating the Network Void: RPC Anatomy in the Distributed Library Catalogue

Scenario:

The university library system allows students to search and reserve books across multiple campus libraries. The catalogue service uses RPC to communicate with the inventory, user account and notification services located on different servers.

- a. (2 marks)** Analyse the components of the RPC anatomy (stubs, runtimes, and network layers) in this library system.
- b. (2 marks)** Explain how these components work together to overcome the physical distance between services.
- c. (2 marks)** Discuss why an Interface Definition Language (IDL) is necessary for this multi-service environment.
- d. (2 marks)** Evaluate the transparency limitations that still exist even with a well-designed RPC layer.
- e. (2 marks)** Recommend how the library system could evolve its RPC usage to better support mobile clients and web frontends.

ETQ 13 (10 Marks): The Paradigm Trade-off: Choosing Communication Models for University Services

Scenario:

The university needs different communication styles: (1) real-time virtual classroom interaction, (2) batch processing of exam results, and (3) administrative updates to student profiles. The architect must choose between synchronous, asynchronous and event-driven models for each.

- a. (2 marks)** Compare Synchronous, Asynchronous, and Event-Driven communication models in terms of coupling, latency and feedback requirements.
- b. (2 marks)** For each of the three university use cases, recommend the most appropriate model and justify based on execution needs and immediacy requirements.
- c. (2 marks)** Analyse the impact of each choice on system scalability and fault tolerance.
- d. (2 marks)** Discuss scenarios where a hybrid approach (combining models) would be beneficial.
- e. (2 marks)** Evaluate the operational complexity introduced by using multiple communication paradigms in one institution.

ETQ 14 (10 Marks): Architectural Defense Mechanisms Against Cascading Faults

Scenario:

During a recent peak registration period, a slow database query in the payment service caused the entire student portal to become unresponsive. The ICT team wants to implement defensive patterns to prevent such cascading failures in future.

- a. (2 marks)** Explain the concept of 'cascading faults' and how they propagate in interconnected university systems.
- b. (2 marks)** Describe how a combination of timeouts, retries with exponential backoff, and circuit breakers forms a robust defense.
- c. (2 marks)** Analyse the configuration trade-offs for each pattern (e.g., timeout values, backoff strategy, circuit breaker thresholds).
- d. (2 marks)** Discuss how these patterns interact when multiple services are involved in a single user request.
- e. (2 marks)** Recommend a monitoring and alerting strategy to detect when these defenses are being triggered.

ETQ 15 (10 Marks): The Event-Driven Blueprint: Apache Kafka as the University Data Pipeline Engine

Scenario:

The university wants a central event streaming platform to capture student activity (logins, course views, assignment submissions), system metrics and integration events from all major applications for analytics, auditing and real-time dashboards.

- a. (2 marks)** Using Apache Kafka as reference, describe the 'engine' of a modern distributed event-driven system for the university.
- b. (2 marks)** Explain how strict guidelines for schema evolution contribute to long-term maintainability and scalability.
- c. (2 marks)** Analyse how non-linear (pub-sub) communication improves fault tolerance compared to point-to-point RPC.

- d. (2 marks)** Discuss the role of topics, partitions, consumer groups and exactly-once semantics in this academic context.
 - e. (2 marks)** Evaluate the operational and governance challenges of introducing Kafka at institutional scale.
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ETQ 16 (10 Marks): The Transparency Dilemma in RPC for Remote University HR Access

Scenario:

The HR department accesses a remote payroll and leave management system via RPC from the main administrative portal. Staff complain that the system sometimes behaves differently from local applications, especially during network congestion or partial failures.

- a. (2 marks)** Analyse why full access transparency cannot be realized in a standard RPC model.
 - b. (2 marks)** Discuss how the introduction of remote reference mechanisms attempts to enhance transparency.
 - c. (2 marks)** Evaluate the remaining visible differences that users (HR staff) still experience.
 - d. (2 marks)** Discuss the trade-off between transparency and performance in this administrative system.
 - e. (2 marks)** Recommend design guidelines the university should follow when exposing RPC-based services to end users.
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ETQ 17 (10 Marks): The Lifecycle of an RPC in a Multi-Service Student Information System

Scenario:

A student portal calls a remote 'getTranscript' procedure on the academic records service. The call involves authentication, database lookup, formatting and notification logging across several internal services.

- a. (2 marks)** Detail the ten-step process of a basic RPC operation from the initial client procedure call to the final return of results.
 - b. (2 marks)** Explain how the involvement of the OS and the network is abstracted away from the developer.
 - c. (2 marks)** Analyse what happens at each layer (stub, runtime, network, server skeleton) during a successful call and during a timeout.
 - d. (2 marks)** Discuss how exceptions and errors are propagated back to the calling code in a distributed environment.
 - e. (2 marks)** Evaluate the debugging challenges that arise because the developer cannot see the full call stack across machines.
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ETQ 18 (10 Marks): RMI vs RPC for Distributed Object Management in Academic Research Tools

Scenario:

A research group has built a distributed simulation tool where Java objects representing experiments and datasets need to be shared and invoked across lab workstations and a central server.

- a. (2 marks)** Compare and contrast Remote Procedure Calls (RPC) and Remote Method Invocation (RMI).
- b. (2 marks)** Focus on the differences in parameter passing and how objects are handled in a distributed environment.

- c. (2 marks) Analyse the role of skeletons versus stubs in each technology.
 - d. (2 marks) Evaluate when RMI would be preferable over plain RPC for this research tool.
 - e. (2 marks) Discuss the language and platform limitations of each approach in a modern heterogeneous university environment.
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ETQ 19 (10 Marks): Evolution of Distributed Communication: From CORBA to gRPC in EdTech

Scenario:

The university's older educational technology platforms were built with CORBA and later WCF. Newer student-facing applications are being developed with gRPC and Protocol Buffers. The architect must understand the evolution and plan integration.

- a. (2 marks) Evaluate the characteristics of CORBA, WCF, and gRPC across multi-language support, performance and complexity.
 - b. (2 marks) Discuss how these technologies have evolved to support modern microservices and cloud-native requirements.
 - c. (2 marks) Analyse the specific advantages gRPC brings for high-throughput student data services.
 - d. (2 marks) Discuss the migration challenges when integrating legacy CORBA/WCF services with new gRPC endpoints.
 - e. (2 marks) Recommend an integration strategy that allows gradual replacement while maintaining service availability.
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ETQ 20 (10 Marks): Comparison of Transient and Persistent Messaging for University Services

Scenario:

The university uses Berkeley sockets (transient) for real-time chat between students and tutors, while IBM MQ (persistent) is used for critical administrative messages such as fee payment confirmations and grade submissions.

- a. (2 marks) Compare and contrast the Berkeley socket interface (transient) with Message-Oriented Middleware (persistent).
 - b. (2 marks) Discuss how the lifecycle of a message differs in each system and the implications for application availability.
 - c. (2 marks) Analyse the reliability and durability guarantees of each approach.
 - d. (2 marks) Evaluate scenarios in the university context where transient messaging is acceptable versus where persistence is mandatory.
 - e. (2 marks) Recommend when the university should prefer one model over the other for new services.
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ETQ 21 (10 Marks): The Role of Abstraction in IBM MQ for Integrating Legacy and Modern University Systems

Scenario:

The university still runs a legacy mainframe for student finance. New web and mobile applications need to interact with it. IBM MQ is used as the integration layer with queue managers, local and remote queues, and logical name resolution.

- a. (2 marks)** Analyse the architecture of IBM MQ, focusing on the relationship between queue managers, local/remote queues, and logical name resolution.
 - b. (2 marks)** Explain how this abstraction simplifies distributed communication for developers of modern applications.
 - c. (2 marks)** Discuss how the queue manager provides location transparency between the mainframe and new services.
 - d. (2 marks)** Evaluate the operational benefits and risks of using a mature MOM product versus newer lightweight brokers.
 - e. (2 marks)** Recommend governance practices for queue naming, security and monitoring in this mixed environment.
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ETQ 22 (10 Marks): Microservices and Event-Driven Architecture for Course Material Updates

Scenario:

When a lecturer uploads new course material in the Learning Management System (LMS), multiple downstream services must react: search index update, student notification, plagiarism check, and analytics pipeline. Direct synchronous calls have caused performance problems.

- a. (2 marks)** Using the example of an 'UpdateCourseMaterial' command, explain how an event bus enables decoupled communication.
 - b. (2 marks)** Discuss the benefits of this approach over traditional synchronous RPC calls in a cloud-native university environment.
 - c. (2 marks)** Analyse the ordering and idempotency requirements for this particular event type.
 - d. (2 marks)** Evaluate the trade-off between eventual consistency and user expectations for immediate feedback.
 - e. (2 marks)** Recommend event schema design and versioning practices for long-term evolution of the LMS ecosystem.
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ETQ 23 (10 Marks): Message Brokerage and Heterogeneity in University Data Integration

Scenario:

Different university departments use incompatible data formats and protocols: the library uses XML, finance uses JSON, HR uses CSV exports from the mainframe, and research groups use custom binary formats. A central message broker is proposed to mediate communication.

- a. (2 marks)** Explain why a message broker is often referred to as an 'Integration Bus'.
- b. (2 marks)** Detail the specific mechanisms it uses to allow disparate applications with different data formats to communicate effectively.
- c. (2 marks)** Analyse the role of message transformation, routing and content enrichment in this heterogeneous environment.
- d. (2 marks)** Discuss the governance and operational challenges of managing many data format mappings.

- e. (2 marks)** Evaluate whether a modern event streaming platform or a traditional enterprise service bus better suits the university's needs.

ETQ 24 (10 Marks): Standard vs Specialized Communication Frameworks for University Workflows

Scenario:

The ICT team is evaluating Apache Camel for complex integration routes versus using Amazon SQS/SNS or a general message broker for departmental workflow automation (e.g., student complaint routing, equipment booking approval chains).

- a. (2 marks)** Evaluate the utility of specialized frameworks like Apache Camel compared to general-purpose queuing services like Amazon SQS.
- b. (2 marks)** Discuss specific development scenarios where a full integration framework provides clear value.
- c. (2 marks)** Analyse the trade-off between development speed, operational complexity and vendor lock-in.
- d. (2 marks)** Recommend criteria the university should use when choosing between these options for different use cases.
- e. (2 marks)** Discuss the skills and training implications for the ICT team under each choice.

ETQ 25 (10 Marks): The Taxonomy of Distributed Failures in the University Wide Area Network

Scenario:

The university operates a multi-campus network with servers in different cities. Network partitions, server crashes, and occasional misbehaving nodes (due to misconfiguration or attacks) have all occurred. The operations team needs a clear failure model.

- a. (2 marks)** Compare and contrast Benign (Clean/Omission) failures with Arbitrary (Malicious/Corrupt) failures in this multi-campus setting.
- b. (2 marks)** Discuss how the defensive architecture required for each type differs, specifically regarding timeouts versus cryptographic defenses.
- c. (2 marks)** Analyse which failure types are most common in a university environment and why.
- d. (2 marks)** Evaluate the cost-benefit of implementing strong cryptographic defenses versus simpler timeout/retry mechanisms.
- e. (2 marks)** Recommend a practical failure model and corresponding defensive layers for the university's critical services.

ETQ 26 (10 Marks): The Impossibility of Perfect Agreement and Its Implications for Timetable Synchronization

Scenario:

Multiple departmental timetable systems must agree on room bookings and exam schedules. Network issues have caused conflicting bookings. The university wants to understand the theoretical limits before investing in stronger consistency mechanisms.

- a. (2 marks)** Analyse the implications of the Two Generals problem for real-world distributed timetable systems.
- b. (2 marks)** Explain how the mathematical impossibility of perfect agreement shapes the way engineers approach system limits and reliability.

- c. (2 marks)** Discuss practical techniques (leases, fencing tokens, eventual consistency with conflict resolution) that can be used despite the impossibility result.
- d. (2 marks)** Evaluate the business impact of occasional conflicting bookings versus the cost of stronger coordination.
- e. (2 marks)** Recommend an acceptable consistency model and user-facing mitigation strategy for timetable conflicts.

ETQ 27 (10 Marks): Strategies for System Survival: Failure Masking in Campus-Wide Services

Scenario:

The university portal, email and library systems must continue operating even when individual servers or network links fail. The team is implementing retransmission, checksums/sequence numbers and replication.

- a. (2 marks)** Evaluate the three primary methods for masking failures — retransmission, checksums/sequence numbers, and replication.
- b. (2 marks)** Explain the specific failure type each method targets.
- c. (2 marks)** Analyse how these methods collectively contribute to 'practical' reliability in a university setting.
- d. (2 marks)** Discuss the overhead (bandwidth, storage, latency) introduced by each technique.
- e. (2 marks)** Recommend a balanced combination of these techniques for the university's most critical student-facing services.

ETQ 28 (10 Marks): Synchronous vs Asynchronous Models and the Asynchronous Dilemma in Real-Time Collaboration

Scenario:

The university's new collaborative document editor and virtual whiteboard require low-latency interaction among students and lecturers. However, the underlying network is asynchronous and clocks are not perfectly synchronized.

- a. (2 marks)** Discuss why timing failures are exclusive to synchronous networks.
- b. (2 marks)** Contrast this with the 'Asynchronous Dilemma' and explain why the lack of perfect clocks makes failure detection an engineering compromise rather than a certainty.
- c. (2 marks)** Analyse the implications for real-time collaborative tools used in teaching.
- d. (2 marks)** Evaluate techniques (vector clocks, operational transformation, CRDTs) that can provide useful guarantees despite asynchrony.
- e. (2 marks)** Recommend an architectural approach for the university's real-time collaboration platform.

ETQ 29 (10 Marks): Designing for Recovery: Building Resilient University IT Infrastructure

Scenario:

Partial failure is the normal state of the university's distributed IT systems. Servers crash, network links flap, and services become slow. The ICT leadership wants to shift the culture from 'prevent all failures' to 'design for recovery'.

- a. (2 marks)** Given that partial failure is the default state of distributed systems, argue for a 'Design for Recovery' philosophy.

- b. (2 marks)** Explain how assumptions like the Fail-Stop model and the use of redundant, identical servers help a system outlive the failure of its individual components.
- c. (2 marks)** Analyse the cultural and process changes required to adopt this philosophy in a traditional university IT department.
- d. (2 marks)** Discuss the role of chaos engineering and regular failure injection in validating recovery mechanisms.
- e. (2 marks)** Recommend concrete practices (circuit breakers, bulkheads, graceful degradation, automated failover) the university should implement.

ETQ 30 (10 Marks): The Security Checkpoint Metaphor: Middleware in the University Web Portal Pipeline

Scenario:

All HTTP requests to the university's main web portal pass through a middleware layer before reaching core application logic. The middleware performs authentication, rate limiting, input validation, logging, compression and response caching.

- a. (2 marks)** Analyse how middleware acts as a series of modular gates protecting the core application logic.
- b. (2 marks)** Discuss the specific roles of logging, authentication, validation, and compression in maintaining 'clean' core application logic.
- c. (2 marks)** Describe the journey of an HTTP request through this middleware-driven architecture, including the importance of the `next()` function.
- d. (2 marks)** Evaluate the strategic value of middleware in standardizing architectural chaos and accelerating CI/CD for university web applications.
- e. (2 marks)** Recommend how the university should structure and version its middleware stack to balance security, performance and developer productivity.

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30 Essay Type Questions for Distributed Systems Theory
Prepared for examination practice and assignment use.